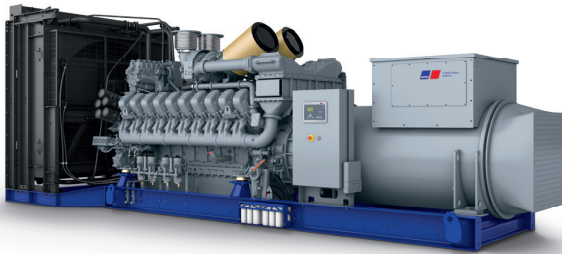




Diesel Generator Set

mtu 20V4000 DS3600

400 V - 11 kV/50 Hz/prime power/NEA (ORDE) + Tier 2 optimized
20V4000G44F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 3370 kVA - 3390 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 75% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Electrical driven radiators
- Medium and oversized voltage alternators
- Low voltage alternator

Emissions

- Tier 2 optimized engine
- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	390
Model		20V4000G44F	Engine jacket water capacity: l	260
Type		4-cycle	Intercooler coolant capacity: l	50
Arrangement		20V	Combustion air requirements	
Displacement: l		95.4	Combustion air volume: m³/s	4.3
Bore: mm		170	Max. air intake restriction: mbar	30
Stroke: mm		210	Cooling/radiator system	
Compression ratio		16.4	Coolant flow rate (HT circuit): m3/hr	80
Rated speed: rpm		1500	Coolant flow rate (LT circuit): m3/hr	44
Engine governor		ECU 9	Heat rejection to coolant: kW (100/110%)	1010/1140
Max power: kWm		2807	Heat radiated to charge air cooling: kW (100/110%)	780/890
Air cleaner		dry	Heat radiated to ambient: kW	105
Fuel system			Fan power for electr. radiator (40°C): kW	105
Maximum fuel lift: m		5	Exhaust system	
Total fuel flow: l/min		27	Exhaust gas temp. (after engine, max.): °C	550
Fuel consumption ²⁾			Exhaust gas temp. (before turbocharger): °C	605
At 100% of power rating:	l/hr	g/kwh	Exhaust gas volume: m3/s	10.0
At 75% of power rating:	674	199	Maximum allowable back pressure: mbar	50
At 50% of power rating:	523	206	Minimum allowable back pressure: mbar	-
	369	218		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized		
		without radiator		
		kWel	kVA*	AMPS
Leroy Somer LSA54.2 ZL17 (LV Leroy Somer standard)	400 V	2696	3370	4864
Leroy Somer LSA54.2 XL11 (Med. volt. Leroy Somer)	11 kV	2704	3380	177
Marathon 1040FDH7103 (Medium volt. marathon)	11 kV	2712	3390	178
Leroy Somer LSA54.2 ZL12 (MV Leroy Somer oversized)	11 kV	2704	3380	177
Marathon 1040FDH7105 (MV marathon oversized)	11 kV	2712	3390	178

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Emission optimized data refer to TA-Luft optimized and NEA (ORDE) optimized/Tier 2 compliant engines.
3 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.